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Abstract

This study explores the effects of two humor styles—pure humor and satire—on audiences' Twitter engagement intentions as well as their attitudes towards mRNA Covid-19 vaccines. In an online between-subjects experiment ($N = 484$), these styles were embedded within a scientist's tweet about the vaccines. The results showed that, compared to a no humor tweet, both pure humor and satire increased audience's expectancy violation, subsequently decreasing both outcomes. Superiority as an individual trait also moderated the relationship between humor styles and both expectancy violation and Twitter engagement. These findings broaden our understanding of different humor styles in science communication on Twitter.

Keywords: humor, satire, expectancy violation, superiority, Twitter

No laughing matter: Exploring the effects of scientists' humor use on Twitter and the moderating role of superiority

As our information environment has increasingly moved towards an online and interactive space, misinformation surrounding salient scientific topics, such as vaccines and climate change, has found new ways to propagate (Allcott et al., 2019). The ramifications of such changes are significant, particularly in the face of the ongoing fight against the coronavirus (Covid-19) pandemic. Vaccine misinformation has fueled skepticism and reluctance, hindering vaccination rates (Kricorian et al., 2021). Although vaccines have been widely available to the American public since May 2021, only a little over 55% of the public had been fully vaccinated by the end of that year (Randall et al., 2021). Similarly, an April 2021 poll from the Kaiser Family Foundation showed that 54% of Americans believed in at least one piece of misinformation about mRNA Covid-19 vaccines (e.g., they can change your DNA) (Hamel et al., 2021). As scientists and science communicators alike work to combat the spread of misinformation, many have turned to directly addressing and refuting misinformation on the same social media platforms (e.g., Twitter) that perpetuate these false claims (Haupt et al., 2021). Such is the context behind our interest in understanding the effects— and implications—of the communication techniques employed by scientists to inform and persuade online audiences.

Of particular interest is the use of humor. In an online space like Twitter, jokes, memes, and other humorous content are often popular forms of communication and have the potential to “go viral,” that is, to reach millions of individuals on these social media platforms (Nahon & Hemsley, 2013). Already, scientists are using humor in their messages on Twitter and Instagram (Cacciatore, 2020; Yeo et al., 2020). These messages from scientists, whether to correct misinformation or to persuade, have the potential to reach a wide audience that otherwise would not be accessible. Still, the effects of humor in science communication, particularly as situated within a social media context, remain understudied: how does humor influence online engagement with these messages, and is it effective as a persuasive tool?

Recent work has already begun to address this question. For example, Marsh (2016) suggests that particular technological infrastructures of social media platforms (e.g., Facebook and Reddit) can differentially influence how users of those platforms respond to humor use in science communication. Still, few studies in the realm of science communication have examined and compared the effects of different *styles* of humor, instead focusing on one particular style or approaching humor as a monolithic technique. But humor styles are similar to communication styles, in that they too can be employed for different purposes and thus affect audiences differently (Berger, 1993; Yuan et al., 2019). As such, understanding the differential effects of humor styles is necessary for a more comprehensive outlook on the overall use of humor in science communication.

In this study, we seek to examine the effects of two different humor styles—pure humor and satire—embedded within Twitter messages about mRNA Covid-19 vaccines in order to understand how audiences engage with these tweets, as well as whether these tweets influence individuals' attitudes towards the vaccines. This study contributes to the current literature by a) exploring expectancy violation as a theoretical explanation for the observed differences between the two styles, b) positing superiority as an individual characteristic that may moderate the relationship between humor styles and engagement and attitudes, and c) expanding our understanding of scientists' humor use on social media. Specifically, we focus on Twitter due to the growing number of scientists (nearly 300,000) who use the platform to share their research, communicate with the public, and network with colleagues in their field (Jünger and Fährnich, 2019; Costas et. al, 2020). Taken together, this study integrates theoretical insights from extant literature on humor, communication styles and science communication to offer insights on the viability of humor as an effective tool for communicating science in an online environment.

Humor Styles

Humor is defined as “anything that people say or do that others perceive as funny, as well as the mental processes that go into creating and perceiving an amusing stimulus, and the emotional response of

mirth involved in the enjoyment of it” (Martin, 2007, p. 5). An individual must first be exposed to a stimulus and perceive that stimulus as playful (thus evoking a non-serious mindset) before they can experience mirth. In 1993, A. A. Berger identified 45 different techniques of humor in his book *The Anatomy of Humor*, classifying the techniques based on their intent and style. The techniques are not mutually exclusive, but Berger noted that different types of humor could be used either separately or simultaneously to satisfy various goals. For example, irony, sarcasm and satire (typically considered “aggressive” or negative forms of humor) can convey social norms in subtle ways, such as by ridiculing out-group members or controlling in-group members. More light-hearted and “pure” forms of humor (like wordplay) can be used as an ingratiation tactic to garner attention or to foster liking (Berger, 1993).

Indeed, recent studies have suggested that scientists are in fact employing different styles of humor in their messages. A 2018 case study that focused on a single science hashtag (#overlyhonestmethods) found over 58,000 relevant, humorous tweets of various styles posted predominately by scientists (Simis-Wilkinson et al., 2018), while a 2022 news article from *Science* explicitly recommended scientists “sprinkle in humor” into their tweets (Brainard, 2022). In another study, researchers have sought to more broadly examine the landscape of humorous content on Twitter and Instagram by conducting a content analysis of all tweets and Instagram posts (within a year-long time frame) that included scientific humor hashtags (Su et al. 2022). Of the more than 110,000 posts extracted, 1200 were randomly selected for manual analysis and coding. The results showed that approximately 40% of the tweets contained wordplay, 33% contained satire, and nearly 75% contained either wordplay, satire or anthropomorphism. Of course, these studies do not provide a definitive estimation for the frequency of *humorous* science tweets within the broader landscape of science tweets; that is, we do not have ample evidence in the literature about the proportional frequency of humorous tweets. However, they do attest to (a) the fact that a growing number of scientists and other individuals have come to see humor as a feature of contemporary online science communication and (b) the need for approaching humor as a technique that can occur through various forms and induce different responses

depending on the particular style. After all, this suggests that even a slight change in how scientists employ humor may have significant implications, as scientists combat potentially deadly misinformation in the face of a global pandemic and other pressing scientific issues. Next, we define our categorizations of the two styles this study focuses on—pure humor and satire—and summarize extant research on the use and effects of these types of humor.

Satire, while widespread in use and appeal, lacks any singular and unifying definition largely due to its diversity in both form and utility (Test, 1991). It is the aggression and judgement towards a particular target offset by a witty and playful approach; it is an act of social critique that seeks to simultaneously cause laughter (Highet, 1972; Test, 1991). Here, we identify satire by two key components—an element of humor and one of aggression—and seek to “create” a satirical tweet. Our approach draws from Skurka et al. (2019), in which the authors similarly defined satire as a combination of humor and indignation and manipulated the presence of one or both of these elements in an existing late night show clip.

Pure humor, then, serves as the counterpart for satire. We define pure humor as any style of humor that induces mirth in audiences, but lacks the aggression and judgment characteristic of satire. Notably, this categorization of “pure humor” is not a value or moral judgment of this humor type but rather is meant to broadly cover any humor technique in which the purpose is “purely,” or exclusively, to elicit laughter and mirth. This definition of pure humor notably does not necessitate the distinction between, for example, wordplay and slapstick humor. Our motivation for using such an umbrella term stems from our interest in the difference between humor styles that are perceived as purely humorous (thus, the category “pure humor”) and those that are perceived as both humorous and aggressive (satire), rather than the subtle distinctions between humor appeals that solely evoke mirth. Focusing specifically on pure humor and satire allows us to clearly isolate the effects of purely humorous appeals and those that incorporate an aggressive, judgmental element. In this study, we also included a “no humor” condition (neither aggressive nor humorous) as a

control. In doing so, we hope to better understand the effects of each experimental condition and to compare the two humor styles against a more traditional and formal style of communication.

Specifically, we are interested in two specific outcomes: tweet engagement and attitudes towards mRNA Covid-19 vaccines. Since the study is situated on Twitter, we conceptualize engagement as particular behaviors unique to the platform—liking, retweeting, and commenting. Our second outcome examines general attitudes towards mRNA Covid-19 vaccines, as the tweets in this study directly address the issue. Next, we outline current literature that has explored the persuasive effects of humor and how humor may influence our outcomes of interest.

Effects of Different Humor Styles

Current findings on both pure humor and satire as strategic forms of communication draw primarily from the health and political communication realms, although there has been a burgeoning focus on science and the environment. Both styles can draw attention to messages, particularly for those who may not be attracted to traditional forms of scientific messaging, and increase the likability of the source (Becker & Waisanen, 2013; Feldman et al., 2011; Yeo et al., 2021). Humor techniques have been proposed to serve as a “gateway” to introduce audiences to other scientific topics, thereby offering even audiences with low prior interest and knowledge the opportunity to engage with various scientific topics (Baum, 2005; Feldman et al., 2011). These findings complement extant qualitative research that has shown that stand-up comedy shows hosted by scientists helped to improve their public images, and the shows were seen as appealing, novel and satisfactory (Bultitude, 2011; Pinto et. al, 2013).

Other studies have found that humor, as a broad persuasive tool, can influence behavioral intentions and attitudes, albeit with mixed findings. On the one hand, humor can produce greater climate change activism intentions (Skurka et al., 2019) and increase the audience's certainty about whether climate change is occurring (Brewer & McKnight, 2015). On the other hand, Riesch (2014) draws from a sociological perspective of humor and argues that since humor often relies on its surrounding social and cultural context,

its use can further divide the ingroup (scientists) from the outgroups (certain audience members). Particularly, humor that ridicules those who do not understand science can further alienate these individuals from science and scientists (Riesch, 2014). Additional research has also supported this idea, showing that satire may lead to motivated reasoning from groups that perceive themselves as the “target” of the joke (Becker, 2014; Feldman, 2017).

Thus, in the context of our study, we hypothesize that satire may backfire against certain groups of individuals and negatively affect their attitudes towards mRNA Covid-19 vaccines. In contrast, a pure humor message that lacks either an explicit target or an aggressive element may have the advantages of general humor techniques without the possible backfire effects. Thus, a pure humor message may positively affect an individual's attitudes towards mRNA Covid-19 vaccines. Moreover, if certain humor types may lead to different behavioral intentions and attitudes, it follows that the audience's engagement with the content (in this case, the tweet) would similarly be different in order to reflect their affective responses or simply to express whether they found the tweet humorous (McAlister, 2014; Yeo et al. 2020). Altogether, we hypothesize:

H1: A pure humor tweet will elicit (a) higher engagement with the tweet and (b) more positive attitudes towards mRNA Covid-19 vaccines, compared to a no humor tweet.

H2: A satire tweet will elicit (a) lower engagement with the tweet and (b) more negative attitudes towards mRNA Covid-19 vaccines, compared to a no humor tweet.

Expectancy Violation

To explore possible explanations for why these differences may arise from different humor styles, we turn to expectancy violation theory (EVT). In EVT, expectancies are defined as “patterns of anticipated behaviors;” that is, we expect individuals to behave and communicate in certain ways due to social norms and other factors (Burgoon, 1993, p. 31). These other factors can include attributes of the communicator (i.e., appearance, gender, age, etc.), the relationship between the communicator and the audience (e.g., their prior

relationship with each other), and the context of their communication (e.g., the environment or situation) (Burgoon & Hale, 1988). Expectancy violations occur when an individual does not behave according to these “expectations.” Ultimately, EVT is an important mediator to consider because it could suggest that individuals hold certain, normative expectations for how scientists should communicate—even on Twitter. That is, potential violations of these expectations could influence how they respond to particular messages (e.g., humorous messages) by scientists on Twitter.

Drawing again from the literature on communication styles, there is reason to suspect that a satire tweet would elicit more expectancy violation than either a no humor or pure humor tweet, and a pure humor tweet would elicit more expectancy violation than a no humor tweet. Both aggressive and polite messages deviate from a traditional, non-humorous message, particularly as one might expect from a scientist. Similarly, both styles of humor used in this study may elicit increased expectancy violation compared to a no humor tweet. More so, aggressive styles of communication may lead to more expectancy violation compared to both polite or neutral styles. In one study, participants who were shown an aggressive letter from a doctor targeting Covid-19 deniers indicated a higher level of expectancy violation (Chu et al., 2021). Another study found that aggressive messages in the childhood vaccination debate led to increased expectancy violation; the communicators of the aggressive message were also seen as more unlikable (Yuan et al., 2019). Since satire combines both humorous and aggressive elements, we expect that satire may have a similar effect as an aggressive style of communication.

H3: Satire will elicit more expectancy violation compared to a pure humor tweet or a no humor tweet.

H4: Pure humor will elicit more expectancy violation compared to a no humor tweet.

Given that different styles of humor can lead to different levels of expectancy violation, the subsequent influences on the outcome variables may also vary. Bullock and Hubner (2020) investigated the implications of political candidates using informal communication on Twitter through the lens of EVT. Specifically, the informal communication condition included puns (i.e., humor), while the formal condition

did not. The authors found that the use of informal communication by political candidates led to an increase in expectancy violation, which decreased the candidate's credibility and lowered the receiver's intentions to support the candidate (Bullock & Hubner, 2020). In the health domain, Yuan et al. (2019) found that blogs using an aggressive tone to target anti-vaccine parents increased expectancy violation. Subsequently, these parents were more likely to view the message as low quality and the writer as unlikable, which contributed to a decreased intention to vaccinate their children (Yuan et al., 2019).

Thus, this suggests a mediation model where EVT could explain the effects of humor styles. Particularly, if the individual perceives the use of either type of humor as unacceptable and wrong (an expectancy violation), they would be less likely to perform either outcome. As such, we propose:

H5: Expectancy violation will mediate the relationship between humor type and tweet engagement.

Specifically, both the pure humor and the satire tweets (compared to the no humor tweet) will lead to increased expectancy violation, which will lead to decreased tweet engagement.

H6: Expectancy violation will mediate the relationship between humor type and attitudes towards mRNA Covid-19 vaccines. Specifically, both the pure humor and the satire tweets (compared to the no humor tweet) will lead to increased expectancy violation, which will lead to more negative attitudes towards vaccines.

Superiority as a Moderator

The mixed findings on humor's effects, discussed previously, suggest a degree of variability in how individuals respond to humorous messages. For example, these mixed—and even null—effects may be explained by a number of individual characteristics; indeed, recent studies have shown that individuals' responses to humor may depend on their sense of humor (Martin & Ford, 2018b) and need for humor (Yeo et al., 2020). Understanding how these factors may differentially influence individuals will allow us to isolate any potential divergent effects and ultimately consider how various humor styles may be more effective for certain individuals than others. In this study, we identify one possible individual characteristic—

superiority—that might further explain these mixed results. In particular, we draw from superiority theory, a classical humor theory that dates back to Greek philosophy (Goldstein & McGhee, 1972), as a foundation for examining the moderating role of superiority on humor effects.

Here, we approach superiority as “a person's arrogant opinion of self as superior to others” (Robbins & Patton, 1985, p. 226). This construct of superiority is an expression of oneself and highlights the “exaggerated self-esteem characteristic” (Robbins & Patton, 1985, p. 226). Additional work on the construct has found that superiority correlates strongly with both attention-seeking and aggrandized self-perception as individual characteristics (Robbins, 1989). These aspects can be linked closely to the ideas and assumptions behind superiority theory, discussed next. For one, the desire to feel superior over another, and its subsequent elicitation of humor, would likely be stronger in those with higher superiority. As such, this construct of superiority is apt for understanding the effects of humor, particularly as rooted in a superiority theory framework.

Superiority theory has traditionally been posited as an overarching theory seeking to explain the effects of humor broadly. Superiority theory examines the relationship between the audience and target of the joke (Berger, 1987). In particular, superiority theory points to a social desire to enhance one's pride and status and to increase self-esteem; thus, mirth is elicited when an individual feels superior over another (Martin & Ford, 2018a). Scholars have argued that the classical version of superiority theory, in an attempt to explain all humor effects, lacks empirical evidence due to its overall vagueness and difficulty to test (Martin & Ford, 2018a). A number of theories based on superiority theory have since been developed, including Zillmann and Cantor's (1996) disposition theory, which states that an individual will appreciate humor more if they hold negative attitudes towards (e.g., feel superior to) the target of the satire (Ferguson & Ford, 2008; Zillmann & Cantor, 1996). In other words, humor effects are contingent on the affective dispositions towards the target of the joke (Zillmann & Cantor, 1996). Similarly, La Fave et al. (1976) argued that humor effects are mediated by the enhancement of one's self-esteem through disparaging different group

or class members. That is, when an individual is exposed to humor that targets others, they may feel superior to those individuals and feel an increase in self-esteem; this felt superiority then influences how they respond to humorous messages (La Fave et al., 1976).

Building on these theories, we argue that superiority as an individual trait can also influence a person's response towards humor. Similar to Greenwood and Isbell (2002), in which individuals with highly sexist and antagonistic attitudes towards women found “dumb blonde” jokes funnier than those with less sexist attitudes, superiority could function in a related manner (Greenwood & Isbell, 2002). That is, someone with high levels of superiority may naturally feel superior towards any general “other.” As such, they may be more likely to appreciate any form of humor in which they perceive another individual as being ridiculed or judged. More so, if we return to the fundamental idea behind superiority theory—that individuals experience mirth as a result of 1) feeling superior over another or 2) wanting to boost one's self-esteem—then it may follow that individuals with high superiority are predisposed to be more receptive towards humor (Ferguson & Ford, 2008; Martin & Ford, 2018a). These individuals, in seeing themselves as generally superior to others, may approach humor (and any style, at that) as a means to reinforce their own superiority. Therefore, they may respond differently than those with lower superiority, who may not view humor in such a way. Again, parallel work from a sociological perspective further urges us to consider the social value of humor. Specifically, if satire is meant to belittle and embarrass particular groups of people, it can both serve to (a) convey social norms and values (subsequently leading to behavior changes) and (b) further divide the in-group from the outgroup (further polarizing groups who do and do not support science) (Berger, 1993; Riesch, 2015). Superiority, then, may serve as an individual trait responsible for exaggerating these effects.

Thus, understanding how superiority operates as a moderator is important for interpreting the immediate and potentially broader effects of both pure humor and satire, revealing how certain groups of individuals might respond positively to these humor styles while others may react more negatively. To our knowledge, no study has examined the moderating effects of superiority as an individual characteristic on

humor effects. Hatzithomas et al. (2021) found that superiority mediated the relationship between exposure to a disparaging but funny advertisement and subsequent attitudes towards the brand (Hatzithomas et al., 2021). However, superiority was conceptualized as identification with the victimizer in the advertisement rather than as a characteristic of the recipient themselves. Other studies on general belief superiority, which captures the extent to which people believe their ideas are superior to others', have shown that increased general belief superiority can lead to an increased likelihood of opinion sharing online and an increased resistance to persuasion (Raimi & Jongman-Sereno, 2020). Thus, given the context of our study, we ask:

RQ2: How will superiority moderate the relationship between humor type and (a) expectancy violation, (b) tweet engagement and (c) attitudes towards mRNA Covid-19 vaccines?

Lastly, while superiority can interact with humor type to influence the two outcomes here, it may also moderate the indirect effects of humor type on the outcomes via expectancy violation. As discussed earlier, those with higher superiority may be generally more receptive to the use of humor in scientists' tweets. Thus, it may follow that these individuals are less likely to perceive either a pure humor or satire tweet as an expectancy violation, compared to individuals with low superiority. Therefore, we ask whether expectancy violation, proposed above as a potential mediator, could explain any moderating effects of superiority. The full proposed model is shown below (Figure 1).

RQ3: Will expectancy violation explain the moderating effects of superiority on the relationship between humor styles and (a) tweet engagement and (b) attitudes towards mRNA Covid-19 vaccines?

[Figure 1 here]

Methods

Participants

We conducted the experiment in June 2021 through the online participant recruitment platform Prolific (Palan & Schitter, 2018) and recruited 500 participants in the United States, who received one dollar for compensation. After excluding those who did not use Twitter or failed to answer at least one of two

attention check questions correctly, we retained 484 participants. The average age of participants was 37.6 years ($SD = 12.1$). Of the sample, 55% were male, 42% were female, 2% were transgender and 1% were other gender types. The majority of participants were White (69%), followed by African American (17%), Asian (6%), Hispanic or Latino (5%) and others (3%). More than half of the participants had received either a bachelor's degree (36%) or a master's degree (24%). The majority of participants reported using Twitter at least several times a week (73%), and 37% who reported that they follow at least one scientist on Twitter.

Experimental Design

After consenting to participate in the study, participants were randomly assigned to one of three conditions, in which they read a tweet from a scientist. The scientist was given a gender-neutral name (Dr. Finley Jones) and profile picture to avoid possible confounding effects of the scientist's gender on the outcome variables. A biography stating "PhD in molecular biosciences | scientist & researcher" was shown to the side of the tweet, as it typically appears on Twitter threads. The purpose of including the biography was to strengthen the salience of the user's scientific background. Finally, social media metrics were held constant across all three conditions at 47 retweets and 586 likes.

The three conditions were *no humor*, *pure humor*, and *satire* (Table 1). Before our main experiment, we conducted two pretests on Prolific. In the first pretest ($N = 127$), we presented a set of eight stimuli to participants, who were asked to rate each tweet based on how humorousness and aggressive they perceived each tweet (1 = *strongly disagree*, 7 = *strongly agree*). For the second pretest ($N = 300$), we selected a subset of candidates from the first pretest that performed closest to our criteria, and randomly assigned participants to one of six tweets. Participants again rated the tweet for humorousness and aggressiveness. Our criteria for each pretest was a set of two tweets that were rated similarly for humorousness, but significantly different for aggressiveness. The two tweets should also differ significantly from the ratings of the "no humor" tweet. The full results and ratings from the pretests, as well as the other stimuli tested, are in the Online Supplement.

[Table 1 here]

In all the conditions, the tweet responds to the common misperception that mRNA Covid-19 vaccines can change an individual's DNA. For the no humor condition, the tweet simply provides the scientific explanation for why the misperception is false. In the pure humor condition, the scientific explanation is presented in the form of a funny analogy, comparing the vaccines to an FBI agent and the virus to a criminal. In the satire condition, the tweet targets and jokingly mocks individuals who believe in the aforementioned misconception while correcting the misperception. Thus, the tweet is designed not only to be humorous but also aggressive towards the target in the tweet. The tweets are held between 235 and 260 characters to be as consistent as possible across conditions.

Prior to exposure to the stimuli, participants responded to questions assessing individual characteristics. After exposure, participants answered questions assessing their responses towards the stimuli, behavioral intentions, and demographic measures.

Key Measures

*Tweet Engagement*¹. Participants were asked how much they agreed or disagreed with three items on a 7-point Likert-type scale: "I would comment on this tweet," "I would like this tweet," and "I would retweet this tweet" (1 = *strongly disagree*, 7 = *strongly agree*). These items were modified from Yeo et al. (2020) and averaged to create an index (Cronbach's $\alpha = .91$, $M = 3.87$, $SD = 1.75$).

Post-Stimuli Vaccine Attitudes. We asked participants to indicate their attitudes towards mRNA Covid-19 vaccines following exposure to the stimuli on five semantic differential scales modified from Lu et al. (2017) on a seven-point scale: *good-bad* (reverse-coded), *harmful-beneficial*, *foolish-wise*, *undesirable-desirable*, and *unreasonable-reasonable*. The five items were averaged to create an index (Cronbach's $\alpha = .96$, $M = 5.58$, $SD = 1.63$).

¹ We have also conducted all subsequent analyses using each item (like, retweet, and comment) as separate dependent variables. The results are included in the Supplemental Materials.

Expectancy Violation. Three items were averaged to create an index, modified from Lu & Yuan (2021). The items, all of which were reverse coded, were as follows: “The scientist's tweet was appropriate for a scientist,” “The scientist wrote in a way I would expect most scientists to write,” and “The scientist used a normal writing style for a scientist.” Participants responded on a 7-point scale of 1 = *strongly disagree* to 7 = *strongly agree* (Cronbach's $\alpha = .87$, $M = 3.74$, $SD = 1.53$).

Superiority. Participants answered nine questions from the superiority scale (Robbins, 1989) on a 7-point Likert-type scale (1 = *strongly disagree* and 7 = *strongly agree*), which were subsequently averaged to create an index. Sample items included “My friends follow my lead” and “I’m witty and charming with others” (Cronbach's $\alpha = .91$, $M = 4.19$, $SD = 1.28$).

Perceived Humorousness and Aggressiveness. On a 7-point Likert-type scale (1 = *strongly disagree* and 7 = *strongly agree*), participants were asked to indicate how humorous and aggressive the tweet seemed. Humorousness was averaged across two items (humorous and funny; $r_{SB} = .87$, $M = 3.45$, $SD = 1.72$) while aggressiveness was also averaged across two items (aggressive and mean; $r_{SB} = .74$, $M = 3.27$, $SD = 1.63$).

Results

Induction Checks

We performed an analysis of variance (ANOVA) to test our manipulations of experimental conditions on the perceived humorousness and aggressiveness of the stimuli. There was a significant main effect of the experimental conditions on perceived aggressiveness, $F(2, 481) = 27.79$, $p < .01$, partial $\eta^2 = .10$. Post hoc comparisons based on Bonferroni correction showed that participants perceived the satirical tweet as more aggressive ($M = 4.02$, $SD = 1.69$) than the pure humor tweet ($M = 2.89$, $SD = 1.50$) and the no humor tweet ($M = 2.92$, $SD = 1.44$), $ps < .001$. There was no significant difference in perceived aggressiveness between the no humor and the pure humor condition. For perceived humorousness, there was also a significant main effect of the experimental conditions, $F(2, 481) = 39.17$, $p < .01$, partial $\eta^2 = .14$. Specifically, the pure humor tweet ($M = 3.83$, $SD = 1.61$) and the satirical tweet ($M = 3.96$, $SD = 1.80$) were perceived as more humorous

than the no humor condition ($M = 2.51$, $SD = 1.35$), $ps < .001$. There was no significant difference in perceived humorousness between the satire and pure humor tweets. As such, the ANOVAs indicated successful manipulations of the satire and pure humor tweets.

Hypothesis Tests

First, we performed an ANOVA to test for main effects of the experimental conditions on three dependent variables—expectancy violation, tweet engagement and attitudes towards the mRNA Covid-19 vaccines. For expectancy violation, there was a significant main effect of the experimental conditions, $F(2, 481) = 30.48$, $p < .01$, partial $\eta^2 = .11$. Post hoc comparisons based on Bonferroni correction showed that the no humor condition elicited a lower amount of expectancy violation ($M = 3.15$, $SD = 1.26$) than the pure humor condition ($M = 3.65$, $SD = 1.39$) and the satire condition ($M = 4.42$, $SD = 1.67$), $ps < .01$. In addition, the pure humor condition also differed significantly from the satire condition in expectancy violation, $p < .001$. These results provide support for H3 and H4. However, there were no main effects of the experimental conditions on either tweet engagement, $F(2, 481) = 0.86$, $p = .43$, partial $\eta^2 = .004$, or vaccine attitudes, $F(2, 481) = 0.48$, $p = .62$, partial $\eta^2 = .002$, addressing H1 and H2.

For subsequent analyses, we used Hayes's PROCESS 4.0 macro for IBM SPSS Statistics (Hayes, 2018). We used Model 4 for our mediation analysis, with the no humor condition as the reference group and a 95% confidence interval (CI) based on 5,000 bootstrapped samples. We predicted that expectancy violation would mediate the relationship between experimental conditions and the two outcome variables, tweet engagement and vaccine attitudes. Our results showed that expectancy violation was a significant mediator for tweet engagement for both the satire condition ($b_{indirect} = -.87$, $SE_{indirect} = .12$, 95% $CI = [-1.13, -.64]$) and pure humor condition ($b_{indirect} = -.34$, $SE_{indirect} = .10$, 95% $CI = [-.56, -.15]$). Expectancy violation was also a significant mediator between vaccine attitudes and both the satire ($b_{indirect} = -.24$, $SE_{indirect} = .08$, 95% $CI = [-.41, -.09]$) and pure humor conditions ($b_{indirect} = -.09$, $SE_{indirect} = .04$, 95% $CI = [-.18, -.02]$). Thus, both the pure humor and satire conditions (compared to the no humor conditions) elicited increased expectancy

violation, which led to a decrease in tweet engagement and more negative attitudes towards mRNA Covid-19 vaccines. As such, we find support for H5 and H6.

Next, we used Model 1 for our moderation analysis, again with the no humor condition as the reference group and a 95% *CI* based on 5,000 bootstrapped samples. We found that superiority was a significant moderator of the relationship between expectancy violation and both pure humor ($b = -0.33$, $SE = 0.12$, $p < 0.01$) and satire ($b = -0.47$, $SE = 0.12$, $p < 0.001$). Specifically, the effects on expectancy violation decreased as levels of superiority increased (Figure 2a). The effects for satire (compared to the no humor condition) were largest for individuals with low ($M-1SD$) superiority ($b = 1.92$, $SE = 0.22$, $p < .001$), then medium (M) levels of superiority ($b = 1.33$, $SE = 0.15$, $p < .001$), and lastly high ($M+1SD$) levels of superiority ($b = 0.73$, $SE = 0.22$, $p < .001$). For the pure humor condition (compared to the no humor condition), the effects again were largest for individuals with low superiority ($b = 0.93$, $SE = 0.21$, $p < .001$), then medium levels of superiority ($b = 0.52$, $SE = 0.15$, $p < .001$), and lastly high levels of superiority ($b = 0.09$, $SE = 0.22$, $p = .66$). For tweet engagement, superiority also moderated the effects of both pure humor ($b = 0.28$, $SE = 0.14$, $p < .05$) and satire ($b = 0.34$, $SE = 0.14$, $p = .01$) (Figure 2b). Individuals with low superiority were more likely to engage with the no humor tweet as opposed to either the pure humor ($b = -0.53$, $SE = 0.25$, $p = .03$) or satire tweet ($b = -0.82$, $SE = 0.25$, $p = .001$). For those with medium superiority, the no humor tweet was likely to have significantly higher engagement compared to the satire tweet ($b = -0.38$, $SE = 0.18$, $p = .04$) but not the pure humor ($b = -0.17$, $SE = 0.18$, $p = .33$) tweet. There were no significant differences in tweet engagement across either the pure humor ($b = 0.18$, $SE = 0.26$, $p = .48$) or satire ($b = 0.06$, $SE = 0.26$, $p = .82$) tweets at high levels of superiority. Lastly, superiority was not found to be a significant moderator of the relationship between vaccine attitudes and either pure humor ($b = -0.03$, $SE = 0.14$, $p = .83$) or satire ($b = -0.06$, $SE = 0.14$, $p = .68$) tweets. Thus, superiority moderated the relationship between the type of humor and expectancy violation (RQ2a) and tweet engagement (RQ2b), but not between humor type and attitudes towards mRNA Covid-19 vaccines (RQ2c).

[Figure 2 here]

Finally, we ran two moderated mediation models using Model 8, with the experimental condition as a multi-categorical nominal independent variable, expectancy violation as the mediator, superiority as the moderator, and either tweet engagement or attitudes towards mRNA Covid-19 vaccines as the dependent variable. For tweet engagement, the indexes of moderated mediation for the pure humor condition and the satire condition were .19 (SE = 0.07, 95% CI = [0.06, 0.33]) and .28 (SE = 0.08, 95% CI = [0.12, 0.43]), respectively. For vaccine attitudes, the indexes of moderated mediation for the pure humor condition and the satire condition were .09 (SE = 0.04, 95% CI = [0.03, 0.17]) and .13 (SE = 0.04, 95% CI = [0.05, 0.22]), respectively. These results show that expectancy violation was a significant mediator for the two-way interaction effects of the experimental condition and superiority on tweet engagement (RQ3a) and vaccine attitude (RQ3b). Our results indicated that for individuals across superiority levels, the satire (vs. no humor) condition violated their expectations for how a scientist typically tweets, which contributed to their decreased tweet engagement and negative attitudes towards the vaccines. This indirect effect was the strongest for those with low superiority levels, weaker for those with medium superiority levels, and weakest for those with high superiority levels. Similarly, the pure humor (vs. no humor) condition also violated participants' expectations for how a scientist should tweet, contributing to their decreased tweet engagement and negative attitudes towards the vaccines. However, this indirect effect was only found among those with low and medium superiority levels, but not among those with high superiority levels. The final model is presented in Figure 3.

[Figure 3 here]

Discussion

With around 300,000 (and likely more) identifiable scientists on Twitter, the social media platform has become a place for scientists to directly communicate with audiences (Costas et al., 2020). Whether to combat misinformation in the face of vaccine skepticism or simply to share their research, scientists are offered the opportunity to engage with the public in a more casual and intimate manner. This study sought to

understand the effects of one such communication strategy—the use of different humor styles (pure humor and satire)—on audience engagement with a tweet as well as subsequent attitudes towards the issue in the tweet (mRNA Covid-19 vaccines). In this study, we explored the relationship between humor styles and the aforementioned outcome variables. First, our results confirmed the role of expectancy violation as a mediator. We also introduced and explored superiority as an individual trait that moderates these relationships. Specifically, we offer an account rooted in a classical theory of humor called superiority theory. In doing so, we expand our understanding of how individual traits can broadly influence our responses to humorous messages. More so, we extend the literature on superiority theory by presenting empirical data on how superiority (as an individual trait derived directly from superiority theory) can account for humor effects. Altogether, our study offers evidence towards the effects of scientists' use of pure humor and satire on Twitter and seeks to present a more thorough perspective on how individuals may respond to different humor styles on social media.

Overall, we found that individuals responded differently in some cases to pure humor, satire and no humor tweets by scientists. Our study focused on two outcome variables—intentions to engage with the tweet and attitudes towards the issue addressed in the tweet (mRNA Covid-19 vaccines). We did not find significant main effects of the experimental conditions on the two outcomes. That is, participants overall indicated similar levels of engagement and attitudes across the no humor, pure humor and satire conditions. Perhaps these behavioral and attitudinal factors are more downstream outcomes that may be too difficult to change with just a single tweet.

However, our results did show that these humor styles indirectly influenced both outcome variables through expectancy violation. Both the pure humor and satire tweets resulted in negative indirect effects through increased expectancy violation. Satire increased expectancy violation, as individuals may not expect scientists to communicate in such an aggressive and judgmental manner. The increased expectancy violation (albeit to a lesser degree) for the pure humor tweet also suggests that individuals do not expect scientists to

communicate in a humorous manner more broadly. Through the increased expectancy violation for both experimental conditions, individuals subsequently expressed lower intentions to engage with the tweet and more negative attitudes towards the vaccines. This suggests that a possible backfire effect occurs when individuals perceive a scientist to be violating their expectations for how the latter should behave (in this case, how they should tweet).

Our results suggest that while Twitter is a social media platform that facilitates casual and humorous communication generally (Pratama, 2020; Wu & Li, 2018), individuals may still expect a scientist to conform to a more formal, traditional and non-humorous style. Thus, when scientists do not communicate through such a manner, individuals are less likely to engage with the tweet or to adopt attitudes compatible with those expressed in the tweet. These findings are consistent with previous studies that have examined expectancy violation in aggressive and polite communication styles (Yuan et al., 2019; Yuan & Lu, 2020), confirming the role of expectancy violation as a mechanism underlying the effects of various humor styles. While the study here cannot answer why individuals may expect scientists to communicate in a more formal, non-humorous manner, future studies can examine whether changing the source of the tweet may influence how audiences respond to these humorous messages. For example, is it more expected of a comedian, or even a layperson, rather than a scientist to engage in these humorous scientific messages, and will audiences be more receptive under those circumstances?

Next, we also investigated the moderating role of superiority as an individual characteristic. We found that those with low superiority levels experienced the highest level of expectancy violation, while the effects were weaker for those with high levels of superiority. Specifically, at higher levels of superiority, individuals no longer differentiated as much in their perceived levels of expectancy violation across the experimental conditions. There are several possible reasons for such a relationship. First, based on superiority theory, those with higher superiority may enjoy humor as a general communication strategy more than those with lower superiority (Berger, 1993; Lintott, 2016). Thus, the relationship between humor styles and expectancy

violation would be weaker (stronger) for those with higher (lower) levels of superiority. Second, individuals with higher levels of superiority may enjoy the tweets targeting and judging others who believe in vaccine misinformation (particularly for the satire condition); these individuals would perhaps perceive themselves as superior to the target in the joke, and express less expectancy violation towards the scientists than those with lower superiority. Thus, when they view a satirical tweet targeting anti-vaxxers from a scientist, their intentions to receive the vaccine may be reinforced while their attitudes towards anti-vaxxers may become more negative. While we did not measure the latter outcome as it was not the focus of this study, we believe that superiority as a measure is inherently related to relevant downstream outcomes, such as increasingly divisive attitudes towards the “out-group” (people who do not believe in science or vaccines), stronger attitudes towards the beliefs advocated by scientists (the in-group), or even a general increase in susceptibility towards humorous persuasive messages by scientists on Twitter.

Additionally, our results also indicated another interesting pattern: those who were low in superiority did not like to engage with either of the two types of humorous tweets regardless. This suggests these types of humorous tweets can also serve to alienate these individuals, perhaps even if they initially hold similar attitudes about the issues. As superiority levels increased, this distinction became less apparent (and insignificant at the highest levels of superiority). In particular, as superiority levels increased, individuals' willingness to engage with any of the three tweets increased. Perhaps then, an individual with high superiority may simply be predisposed to engage with a tweet regardless of the style of humor used, as this reflects their desire to express their opinions so others can admire them or follow their lead (Robbins, 1989). Interestingly, our results also indicated the differences between the pure humor and no humor conditions started to diminish at medium levels of superiority, while the differences between the satire and no humor conditions only began to diminish at high levels of superiority. While we cannot conclusively determine why superiority seems to have a stronger effect on the pure humor condition than the satire condition, perhaps the difference between the pure humor and the no humor conditions was less than that of the difference between the satire and no

humor condition from the onset. In other words, the pure humor tweet only differed from the no humor tweet in terms of humorousness (and not aggressiveness). The satirical tweet was different from the no humor tweet for both humorousness and aggressiveness. Thus, this added element of aggressiveness may have contributed to this underlying difference between the pure humor and satire tweets.

Finally, while we did not find moderation effects for attitude, the moderated mediation analyses identified expectancy violation to mediate the indirect effect of the experimental conditions on both attitude and tweet engagement, moderated by superiority. That is, as an individual's superiority levels increased, their perceived expectancy violation decreased. Subsequently, they expressed tweet engagement intentions and more positive attitudes towards the vaccines. Returning to the earlier discussion, this may occur since those with higher superiority appreciate humor, regardless of the source, more than those with lower superiority. Thus, they do not perceive a scientist tweeting with humor as a phenomenon that would particularly violate their expectations. These individuals would respond positively towards the tweet and the scientist, including engaging with the tweet and changing their attitudes accordingly.

The social and practical implications of such are significant. Broadly, the results suggest there are set normative expectations for how scientists should communicate on Twitter, particularly with regards to the use of humor. So, while Twitter does offer scientists a platform to communicate in a more direct and unfiltered manner, this study suggests that, at least for scientists, the norms of the social media platform may be superseded by the expected norms of general science communication. Moreover, these expectations are held strongest by those with the lowest levels of superiority, such that the use of certain humor styles may actually alienate these individuals. Thus, this study raises important future research questions: what exactly are these expectations, and how do they hold across different social media platforms? As scientists explore novel persuasive techniques, such as humor, how does the use of these tools fit within these expectations?

Altogether, the results show that humor (and its various styles) should not be approached as a technique towards which all audiences will be equally responsive. In this study, we focused on one particular

trait—superiority. Specifically, we used the superiority scale from Robbins (1989), which was derived from Kohut's (1971) psychoanalytic theory of the self. Similar to other personality and trait scales, the superiority scale consists of nine self-reported items used to assess “a person's arrogant opinion of self as superior to others” (Robbins & Patton, 1985, p. 226). Individuals with higher levels of superiority may often be perceived as socially gregarious and narcissistic, with subsequent studies identifying superiority as a component of the Narcissistic Personality Inventory (Raskin & Terry, 1988; Robbins, 1989). Importantly, narcissism has been linked to increases in the time spent on social media as well as in the likelihood of being opinion leaders on social media (Song et al., 2017; McCain & Campbell, 2018). These imply that individuals high in superiority may also share these tendencies, and as such, are a key demographic to be targeted by scientists and science communicators. Certainly, the concept of “superiority” is abstract and necessarily confined to the particular measurement used in this study; however, it (a) indicates that even the natural variations in whether we see ourselves as superior to others can temper our responses to humor and influence our behaviors and (b) provides empirical support for a key humor theory, which may subsequently inform future humor studies in science communication.

Arguably, the concept is also related to other variables of interest in science communication, which could be tested in future studies. For example, levels of superiority may influence the strength or rigidity of an individual's pre-conceived scientific attitudes, such that the attitudes of individuals with higher superiority may be more difficult to change. This may suggest that the use of satire and pure humor in tweets is a relevant and important messaging strategy for changing the attitudes and beliefs of this particular subpopulation, for whom traditional messages may not be as effective and persuasive. Similarly, superiority could influence the degree to which an individual trusts scientists or would prefer to trust their own research, or even interact with political ideology to influence whether they perceive scientists as a (superior) in-group or an (inferior) out-group member.

Moreover, other audience characteristics could also influence how people respond to humor, such as the big five personality traits. For example, extraversion, which describes how sociable and energetic an individual is, could moderate levels of engagement with humorous content, while individuals with high agreeableness may prefer tweets with pure humor over tweets containing satire (John and Srivastava, 1999). Also, as this study was conducted in the U.S., it would be worthwhile to examine whether these effects hold across different cultural contexts. While research has generally confirmed that certain cognitive processes underlying humor responses are universal (Alden et. al, 1993), individuals from other countries may hold different normative expectations about how and whether scientists should use humor in their communication. As such, future research should seek to better understand how different individuals or groups may respond differentially to the use of humor in science communication more broadly.

Certainly, there are several limitations to this study. First, while the pure humor and satire conditions were both rated as more humorous than the no humor condition, the objective ratings of both were just around the midpoint on the 7-point Likert scale. That is, participants did not find the tweets to be particularly funny by themselves, only more so than the no humor condition. Similarly for aggressiveness, the satire condition was rated near the midpoint, albeit a few points higher than the pure humor and no humor conditions. Second, while we did base our experimental conditions on real tweets, only a few words differed across each condition to indicate the particular manipulation. These subtle changes in wording may not have been obvious enough to elicit larger differences across conditions. To address these first two limitations, future studies will have to carefully select or create stimuli that mimic real-world viral and humorous messages while tightly controlling for extraneous variables that might influence the results otherwise. Additionally, future studies can explore tweets that induce stronger levels of humorousness and aggressiveness than those in this study to assess whether our findings can be replicated with different pure humor or satire tweets. Third, each participant was exposed to a single message in our experiment. The single-message exposure raises the question of how individuals might react if they are frequently exposed to such humorous messaging from

scientists. Or, if such a message were to appear by itself within an individual's larger Twitter feed, might the observed effects be even smaller? Fourth, our sample was drawn from a population of Prolific users, which limits the generalizability of this study. So too, the effects of the experiment may be limited to the particular platform (Twitter) and context (mRNA Covid-19 vaccines). Only 22% of the U.S. population use Twitter; as such, their behaviors and preferences may not reflect that of the broader population at large (Hughes & Wojcik, 2019). Regarding the context chosen for this study, participants may have already developed strong attitudes towards mRNA Covid-19 vaccines due to their politicized nature or other factors (Druckman et al., 2021); this could also explain why we did not observe any main effects on attitudes. Thus, future scholarship could examine the effects of humor styles embedded in a different social media platform, or to explore a different scientific context that may be less controversial or prominent at the time of investigation.

Conclusion

In summary, the present study offers theoretical and empirical insights on humor styles in science communication. Our findings present a more nuanced perspective for the practical use of pure humor and satire by scientists on Twitter. While Twitter is a space where humorous content and messages often “go viral” for their appeal, our study suggests that science communicators should be cognizant of potential backfire effects that may arise from their use of various humor styles. While individuals on Twitter may be receptive towards such messages from other Twitter users, they may still expect scientists to communicate in a more straightforward, neutral and non- humorous way. Taken together, this study 1) highlights the need to better understand the effects of humor in science communication, particularly as it occurs on various social media platforms and 2) suggests that science communicators carefully evaluate how they use various forms of humor to communicate with their audiences on these platforms.

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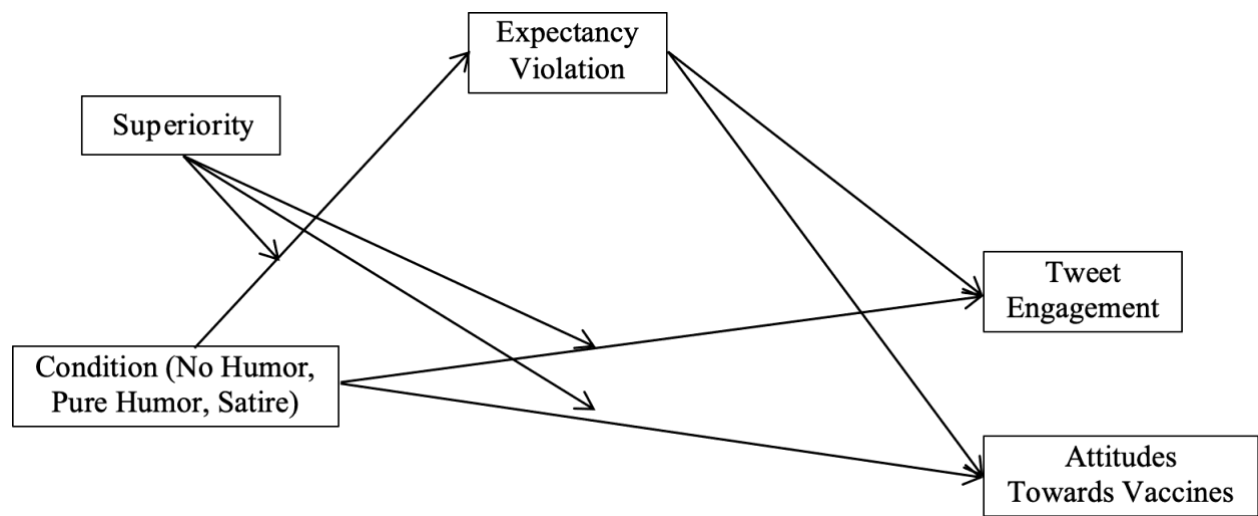


Figure 1. Proposed model.

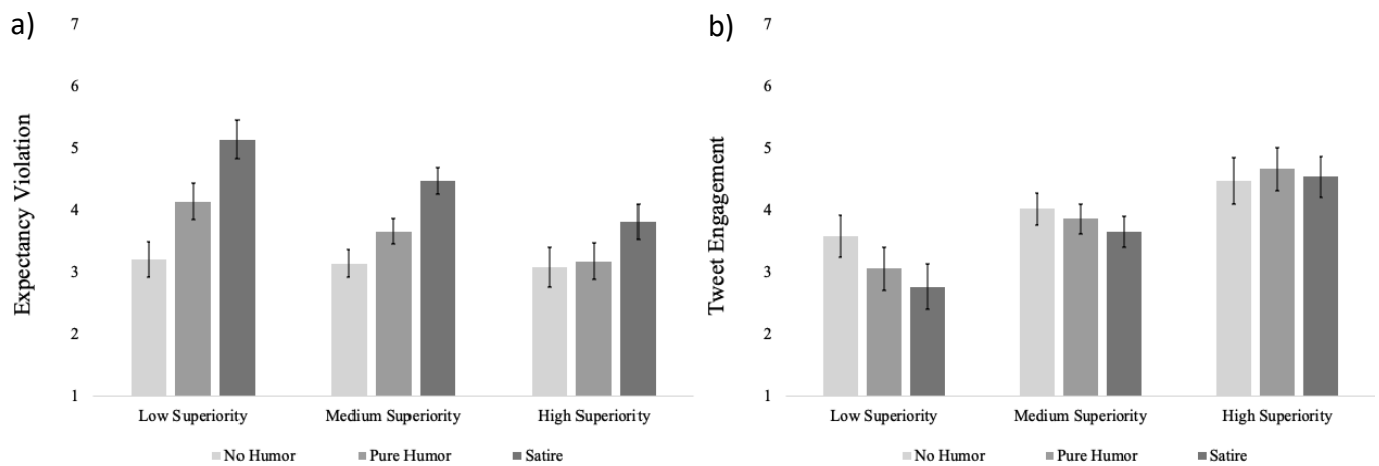


Figure 2. Moderating effect of superiority on experimental conditions (no humor, pure humor or satire) on a) expectancy violation and b) tweet engagement. Low superiority represents one standard deviation below the mean, medium represents the mean, and high represents one standard deviation above the mean. Error bars represent 95% confidence intervals.

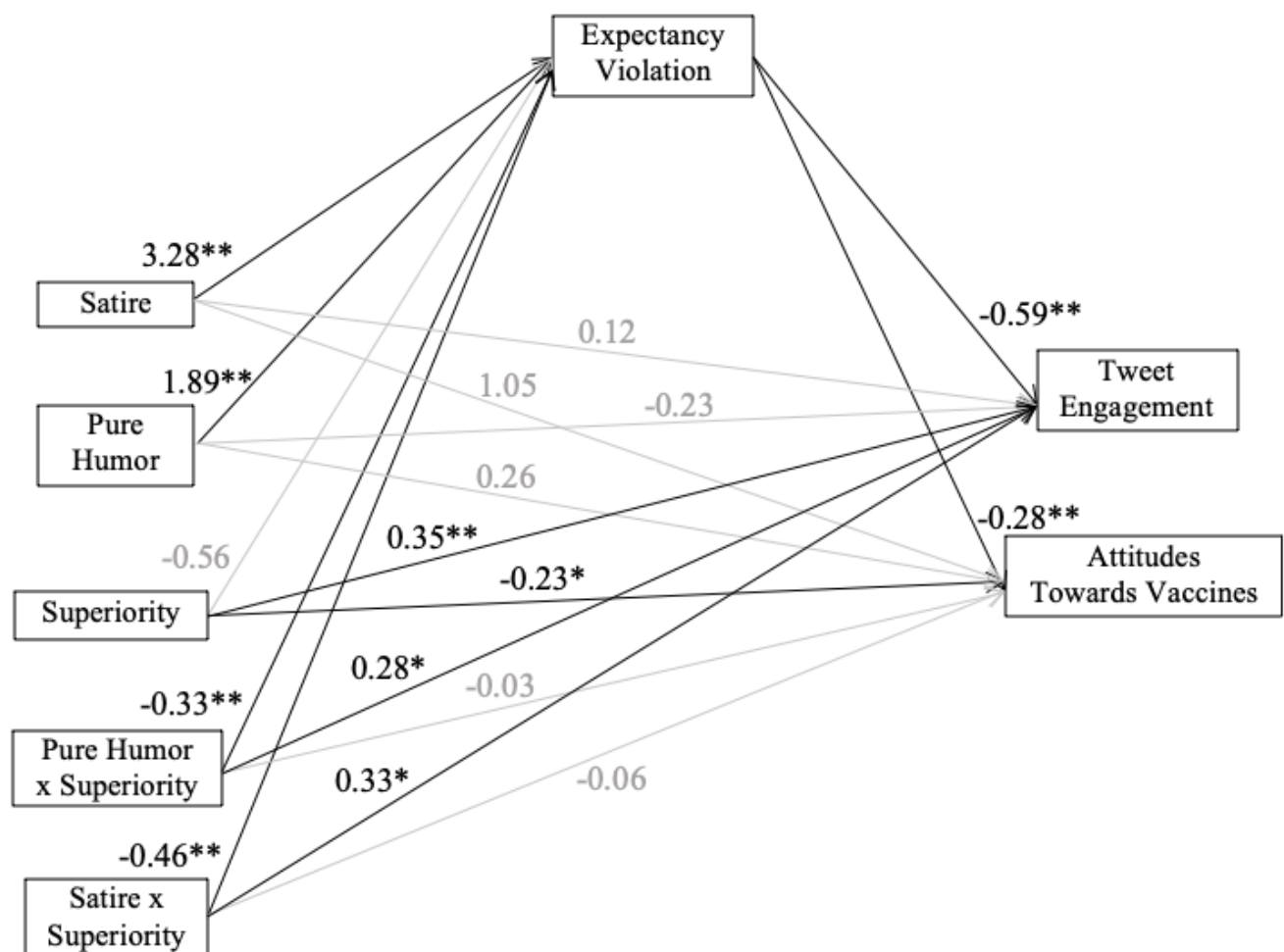


Figure 3. Full model with unstandardized coefficients for the conditions; * $p < 0.05$
 ** $p < 0.01$; gray lines and coefficients are nonsignificant.

Table 1. Stimuli chosen for each condition.**No Humor Tweet:****Dr. Finley Jones, PhD.**

@drfinleyjones

"The COVID mRNA vaccine will change your DNA!" No, it won't. It does not have the ingredients nor ability to do that, and it is synthetic. The vaccine sends instructions for your body to make the protective response and then disappears.

7:06 PM - 27 May 2021

47 Retweet 586 Likes

**Relevant People**

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**Dr. Finley Jones, PhD.**

@drfinleyjones

PhD in molecular biosciences | scientist & researcher

Pure Humor Tweet:**Dr. Finley Jones, PhD.**

@drfinleyjones

"The COVID mRNA vaccine will change your DNA!" No, it won't. You can think of it as the FBI agent who creates the "Most Wanted" profiles. He'll recruit other parts of your body to make sure the next time they see the criminal, he's...well, handled "accordingly."

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Satire Tweet:**Dr. Finley Jones, PhD.**

@drfinleyjones

"The COVID mRNA vaccine will change your DNA!" These people are so afraid that they'll turn into trees or something...which would be an improvement for them anyways. The vaccine only sends instructions for your body to make a protective response, and then disappears.

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